Faculty of Engineering
Ain Shams University.
Mechanical Power Department

Simulation of Low Reynolds Number UAV
ICMTC Competition



By: Ahmed Reda Elshazly
Phone Number: +201020805760
Linkedin: Link
Email: ahmed.shazly.student@gmail.com

Table of Contents

Nomenclature 2

Objectives 2

UAV Design Specs 2

Geometry Generation 4

Inflation layer and boundary layer calculations 6

1. Reynolds Number (Re) 6
2. Skin Friction Coefficient (Cf) 6
3. Wall Shear Stress (τw) and Friction Velocity (uτ) 6
4. First Layer Height (Δy) for y+=1 6
5. Boundary Layer Thickness for Turbulent Flow 6

Solver Settings and Boundary Conditions 7

Settings 7

Recommendation for using GEKO 7

Input and Output Parameters 8

Convergence criterion 9

Mesh Dependency Test 9

Mesh Visuals 10

Solution and Post Processing 12

Flow Inspection 12

Stall Angle 12

Post Stall Angle 14

Cruise 15

Lift, Drag and Pitching Moment Polar 16

1. Lift Curve 16
2. Drag Polar 16
3. Pitching Moment 17
4. CL-CD Curve 18

Spanwise Lift Distribution 18

Summary of Key Aerodynamic Factor 20

Nomenclature

•δ_e: Elevator deflection

•VLM: Vortex Lattice Method

•C_L: Lift Coefficient

•e: Oswald Efficiency Factor

•AR: Aspect Ratio

•C_m: Pitching Moment

•C_D Drag coefficient

•S_{ref}: Wing's Reference Projected Area

•C_{VT}: Vertical Tail Volume Coefficient

•C_{HT}: Horizontal Tail Volume Coefficient

•$L_{Fuselage}$: Fuselage length

•λ: Taper Ratio

•$\bar{C}$: Mean Aerodynamic Chord

•α/AoA: Angle of Attack

• C_{Lmax}: Maximum 3D Lift Coefficient

•C_{lmax}: Maximum 2D Lift Coefficient

•$C_{L\alpha}$: Lift Curve Slope

•$C_{m\alpha}$: Pitching Moment Slope

•C_{sep}: GEKO Separation Free Parameter

•C_{mix}: GEKO Mixing Free Parameter

Objectives

This Report documents the process of validating the design of a Low speed Reynolds Number propeller UAV we designed at Aero ASU for participation in ICMTC UAVC competition. The design features a wwith MTOW of 10 Kg, that roughly makes the stall speed of the aircraft at 11 m/sec with estimated cruise speed of 16 m/sec. the requirements in this study is:

•Perform a parametric study to get the Lift, Drag and Pitching Moment polar at the cruise speed of 16 m/sec with **trimmed configuration, and verify that the elevator deflection is sufficient to get CM=0 at α=3**.

•Extract the cruise speed **Oswald efficiency factor e**.

•Find the Maximum **CL/CD** for our Range focused Design.

•Compare simulation results with preliminary calculations and VLM simulations.

•Plot the **spanwise lift distribution** and compare it with the ideal case.

UAV Design Specs

Design Parameter	Value
Mass (Kg)	10
X_{CG}	0.86
S_{ref} (m^2)	1.104

$\bar{C}\ (m)$	0.47
$Aspect\ Ratio$	5
λ	1
$L_{fuselage}\ (m)$	2.2
C_{VT}	0.1
C_{HT}	0.5
Cruise Speed (m/sec)	16
δ_e **(deg)**	-3.5 (deflected up)
Wing Airfoil	SG6043
Tail Airfoil	NACA0009

Geometry Generation

The UAV was modeled in SolidWorks 2025 and the Computational Domain in SpaceClaim, to capture the gap effect of the control surfaces and flaps, the recommended gap is less than 1% of the chord, to prevent the compression of boundary layer in the gaps, a gap of 3mm were made.



Figure 1:UAV model in Space claim

To clean up the cad model, fillets were made for the landing gear and the trailing edge of the lifting surfaces to prevent sharp angle transition.

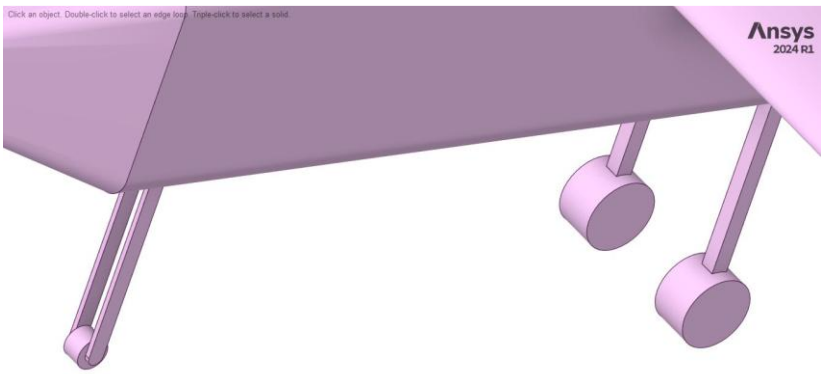


Figure 2:Landing gear before cleaning

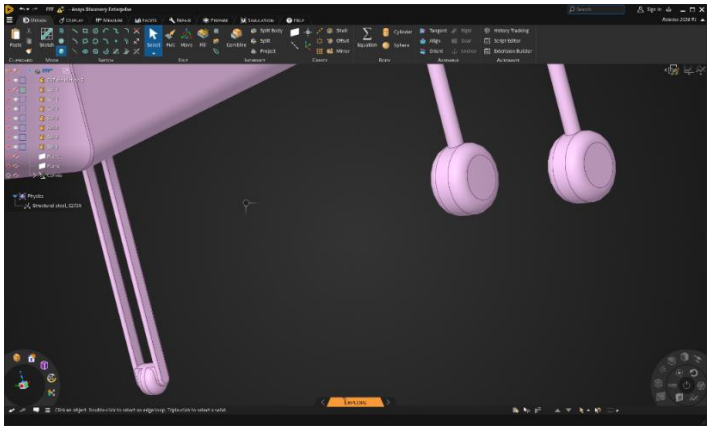


Figure 3:Filleting the landing gear and close problematic gap in the nose landing gear.

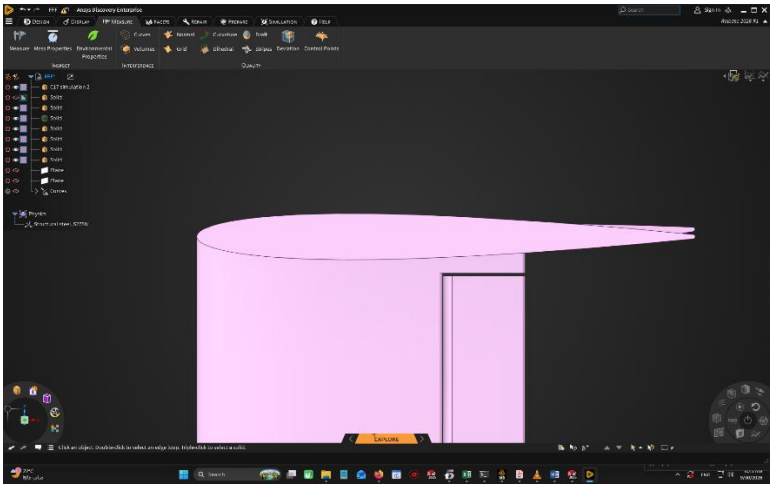


Figure 4:Deflected Elevator

A custom cone like enclosure was made to decrease the number of cells.

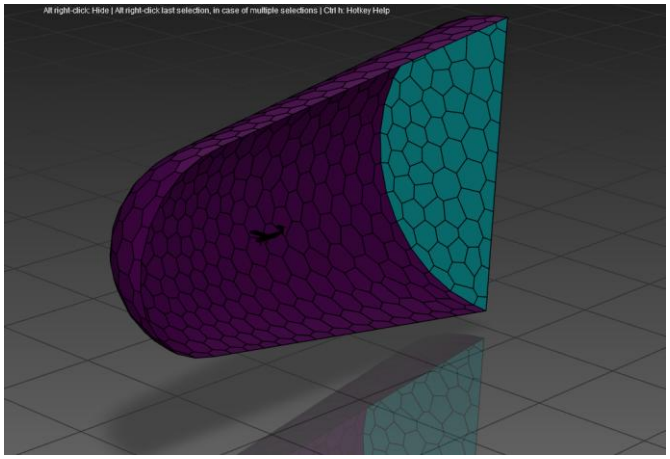


Figure 5:Computational Domain

Inflation layer and boundary layer calculations

To estimate the smallest cell height possible, we made calculations based on the aircrafts maximum speed which is 20m/sec, and we used properties of air at sea level.

1. Reynolds Number (Re)

Using U=16 m/s, L=0.47 m, and ν=1.5×10−5 m2/s:

$$Re = \frac{UL}{\nu} = \frac{16 \cdot 0.47}{1.5 \times 10^{-5}} = 501333$$

2. Skin Friction Coefficient (Cf)

Using Schlichting's empirical formula for turbulent flat plates:

$$Cf = [2log(501333) - 0.65]^{-2.3} \approx 0.00424$$

3. Wall Shear Stress (τw) and Friction Velocity (uτ)

$$\tau w = \frac{1}{2} \cdot C_f \cdot \rho \cdot U^2 = 0.5 \cdot 0.00424 \cdot 1.225 \cdot 16^2 \approx 0.665\ Pa$$

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} = \sqrt{\frac{0.665}{1.225}} \approx 0.7368\ m/s$$

4. First Layer Height (Δy) for y+=1

$$\Delta y = \frac{\nu \cdot y_+}{u_\tau} = \frac{1.5 \times 10^{-5} \cdot 1}{0.7368} \approx 2.03 \times 10^{-5}\ m$$

5. Boundary Layer Thickness for Turbulent Flow

$$BL = \frac{0.37 \cdot C}{Re^{\frac{1}{5}}} = \frac{0.37 \cdot 0.47}{501333^{\frac{1}{5}}} = 0.013m$$

Solver Settings and Boundary Conditions

Settings

•Solver: Pressure Based Steady State

•Turbulence Model: **K-Omega GEKO** with **Csep=1.75**, **Cmix=0.35**. with **Curvature Correction** and **Algebraic Transition** enabled.

•Velocity Inlet: Magnitude and direction, with velocity **U and angle of attack AoA as input parameter**

•Outlet: Pressure gauge pressure=0

• **Fluent GPU Solver** was used to speed up the **parametric study** calculations, **SIMPLEC** was used along with second order upwind scheme and local pseudo transient time step, all other settings were left untouched.

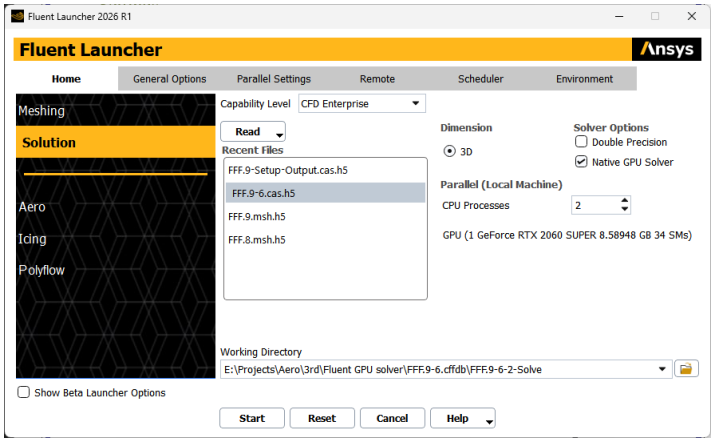


Figure 6:Fluent GPU Solver Enabled

Recommendation for using GEKO

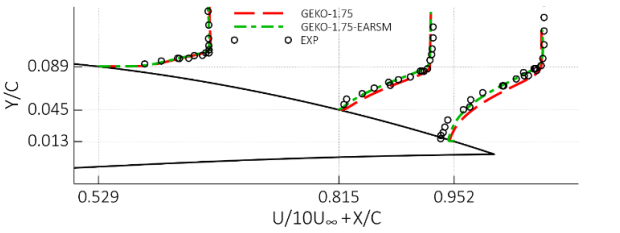


Figure 29: NACA-4412 airfoil near the trailing edge with streamwise velocity profiles at measurement locations predicted with linear and non-linear GEKO-1.75 model compared with the experimental data [38].

Figure 7: Menter Experiment with GEKO.

GEKO was recommended by Florian Menter at ANSYS for low Reynolds numbers airfoils like NACA4412, comparing his simulation results with NASA's experimental data.

Input and Output Parameters

Created input parameters to be able to conduct parametric study in Fluent, varying AoA and outputting aerodynamic coefficients

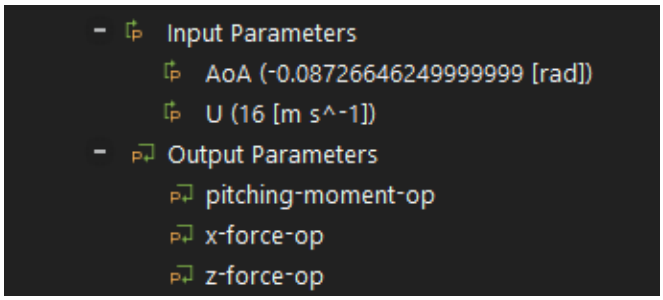


Figure 8:Input and Output Parameters

Lift, drag and are calculated as follows:

$$C_L = -C_x \cdot \sin(\alpha) + C_Z \cdot \cos(\alpha))$$

$$C_D = C_x \cdot \cos(\alpha) + C_Z \cdot \sin(\alpha)$$

Then we get the lift and drag coefficients by dividing by $\frac{1}{2} * \rho * U^2 * S_{ref}$ where $S_{ref} = 0.552\ m^2$ "Half the projected area of the wing"

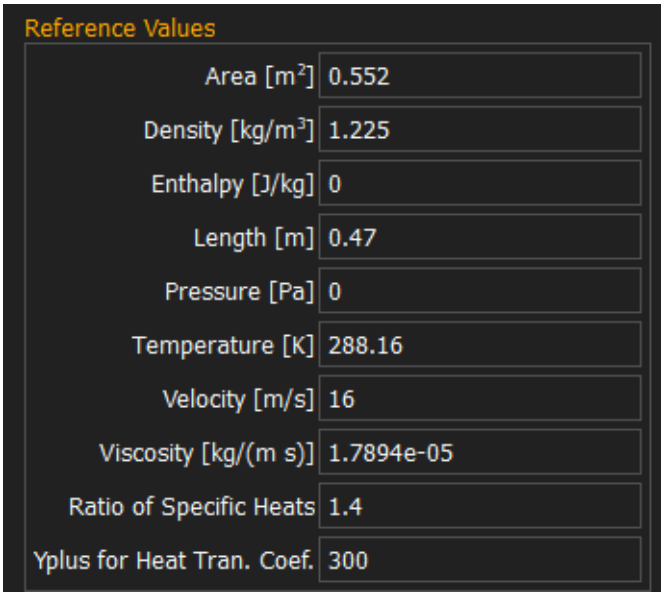


Figure 9:Reference Values

Convergence criterion

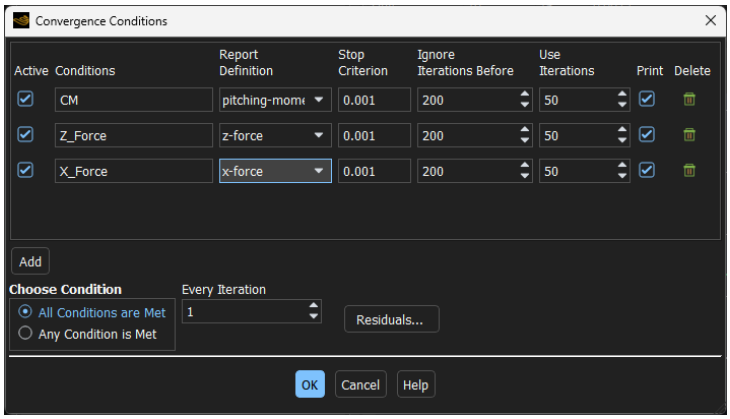
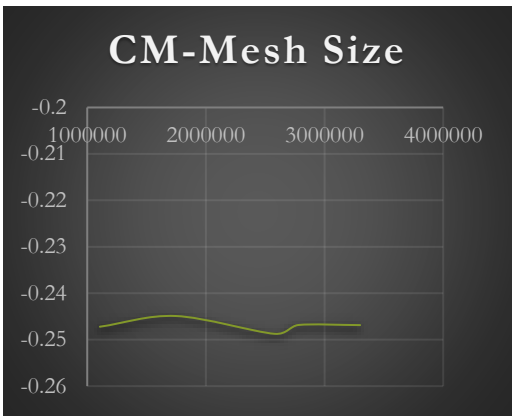
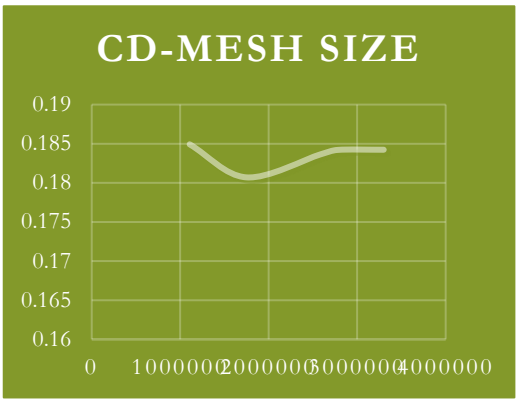
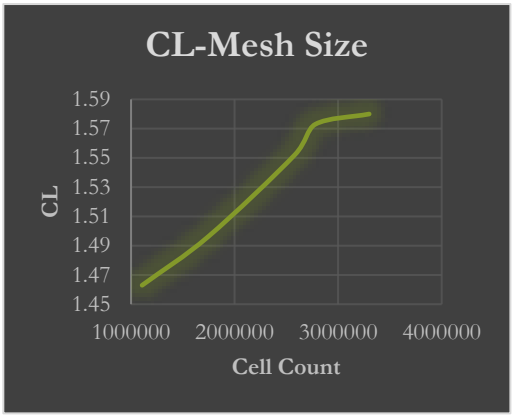


Figure 10: Convergence criterion

Mesh Dependency Test







•The test was conducted at AoA=16 degrees, since the mesh passed this high complex problem, all lower AoA's will be mesh independent.

•Drag and pitching moments did not change much, but the lift coefficient needed more cells to be mesh independent., thus the 2.78 million cells grid was selected.

•2 Inflation Layers settings were used, one for the whole aircraft. And thinner one on the gaps to squeeze as much inflation layers as possible.

Mesh Visuals

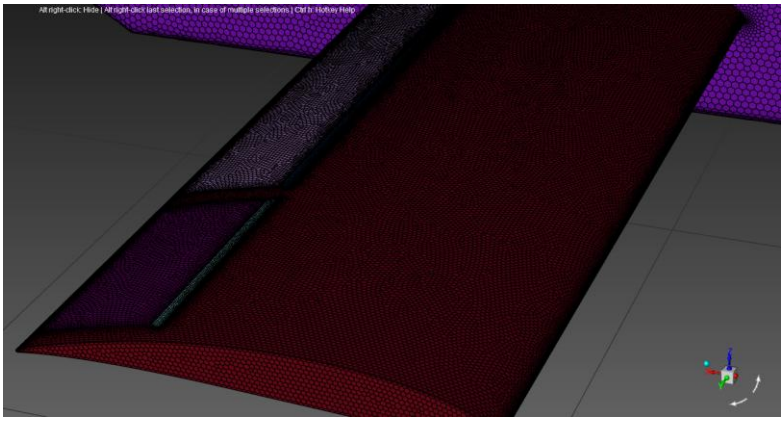


Figure 11: Wing's mesh

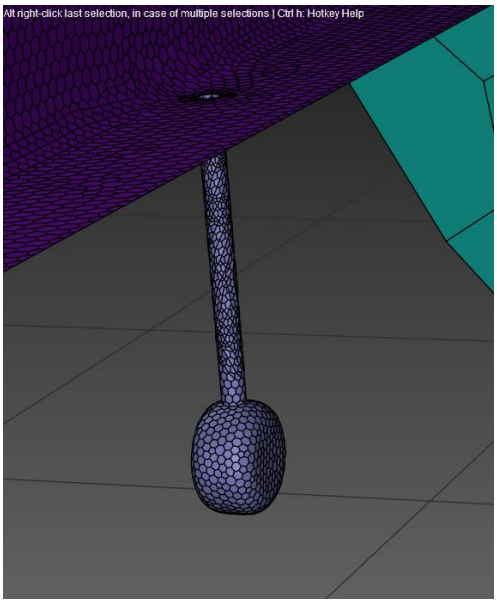


Figure 12: Landing Gear Mesh

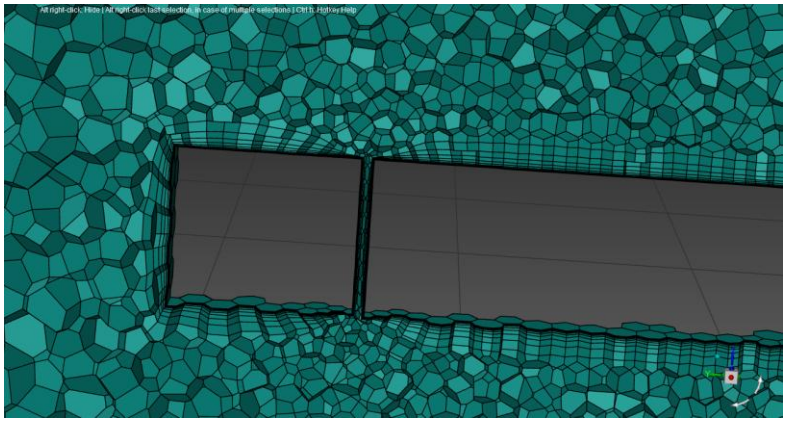


Figure 13: Wingtip with airleron gap.

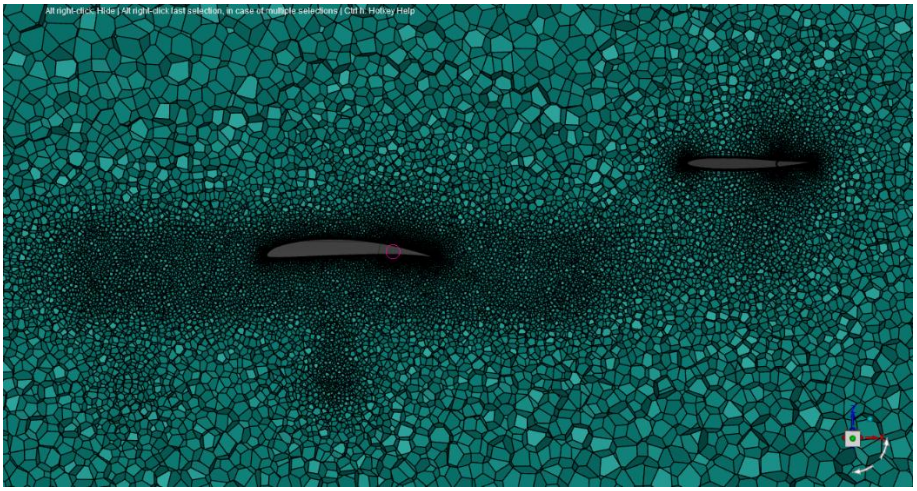


Figure 14: Section plane for wing and tail.

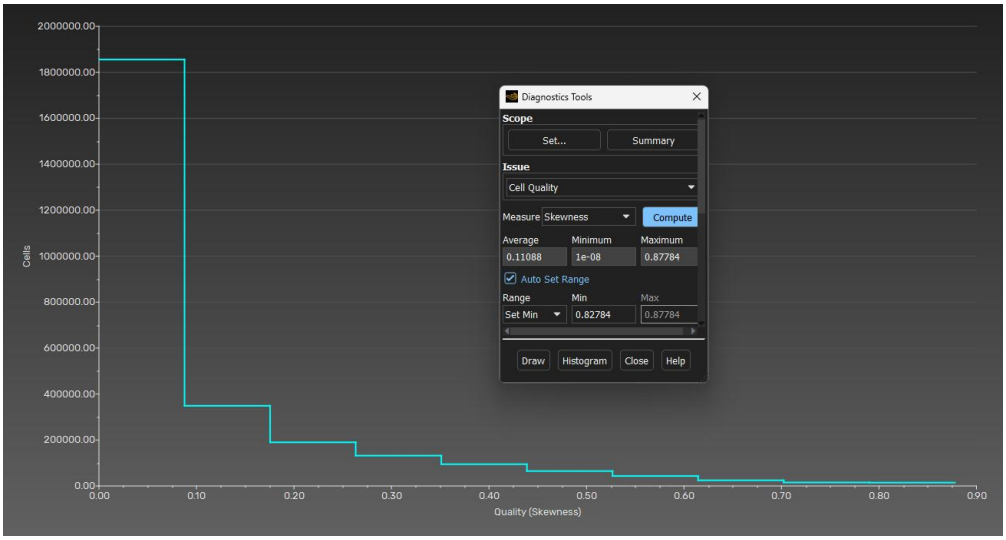


Figure 15: Maximum Skewness is 0.878

Solution and Post Processing

• A sweep of angle of attacks from 18 to -5 degrees were simulated.

It is clear that the stall happened at 17 degrees, followed by a rapid lift reduction, we will be investigating the flow characteristics at α_{cruise}=2, α_{stall}=17 and $\alpha_{post\ stall}$=18

Flow Inspection

Stall Angle

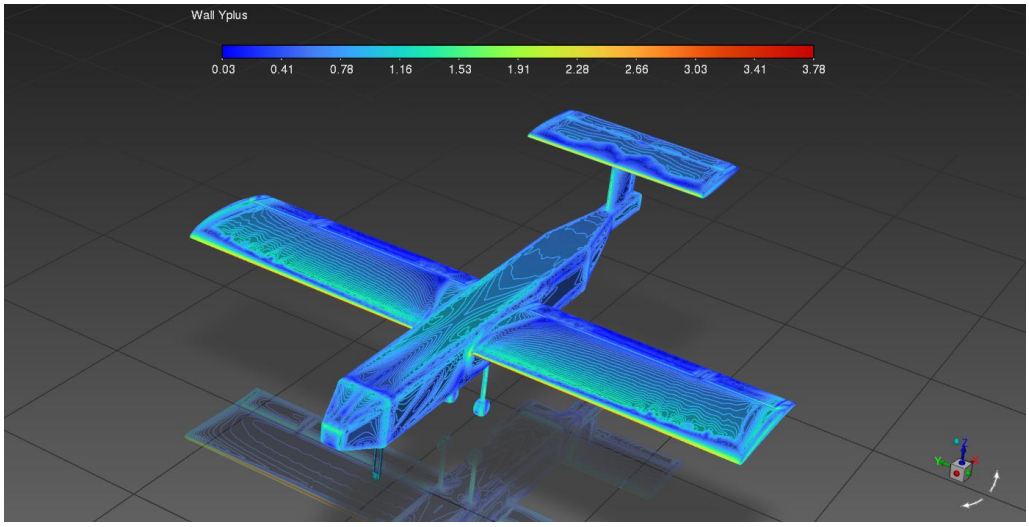


Figure 16: y+ contour a=17

The y+ contour shows sub 5 values, but it drops below 1 at the suction peak and it drops again at the trailing edge.

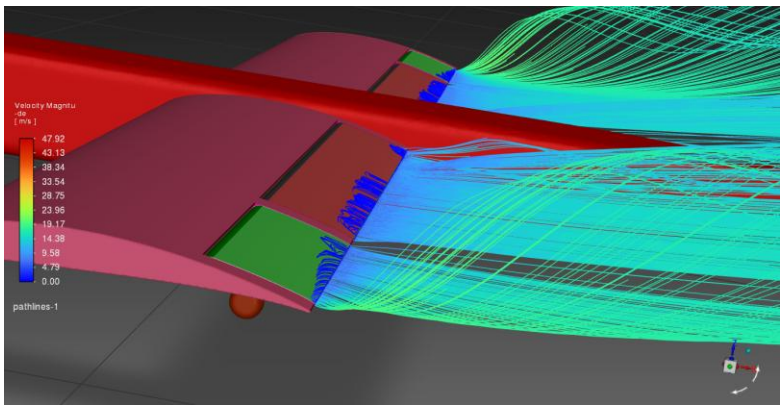


Figure 17:Trailing edge pathlines

The pathlines at the trailing edge shows that the high-pressure air rolls up at the trailing edge to the lower pressure zone, this is because a round trailing edge makes it much easier for the air to roll up to the upper side, while a blunt trailing edge acts as a geometric anchor for separation, because it satisfies the Runge-Kutta condition, it also accelerates convergence due to less oscillation. This behavior increases drag can accelerate stall because of the additional adverse pressure gradient.

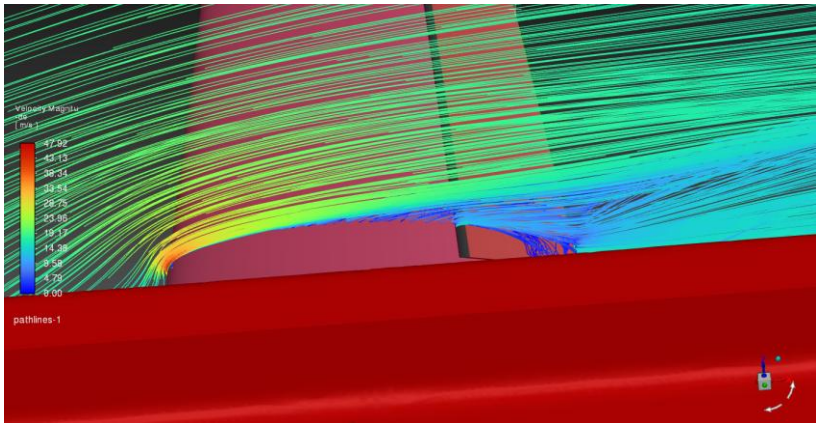


Figure 18: Plane Pathlines

A plane pathlines shows it clear that there's indeed separation happening near the leading edge, which then gets reattached by the high-pressure air rushing through the hinge line energizing the boundary layer, then again adverse pressure gradient from the flow rolling up at the trailing edge causes it to separate again, this would give false feeling of being away from the stall, then the sudden drop of lift that occurred at α=18.

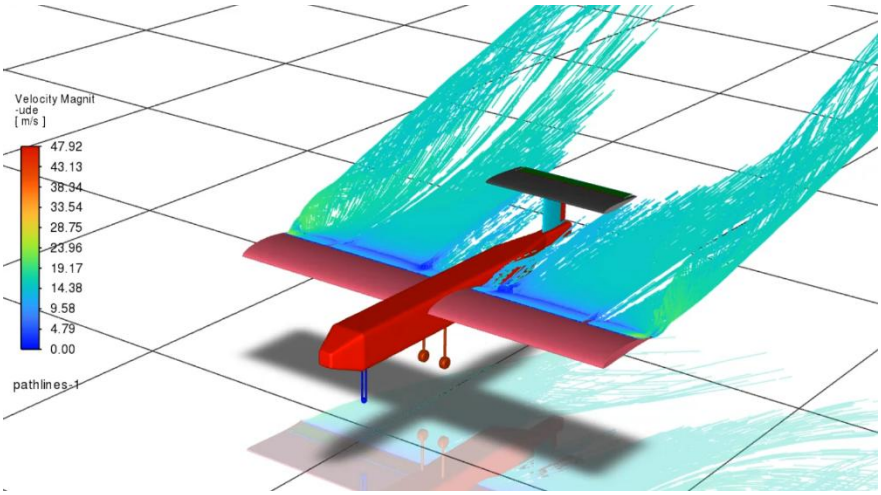


Figure 19:Tip Vortices

Strong Tip vortices can be seen in this pathlines, with the appearance of the separation region.

Here's a **Video Animation** of the tip vortices.

Post Stall Angle

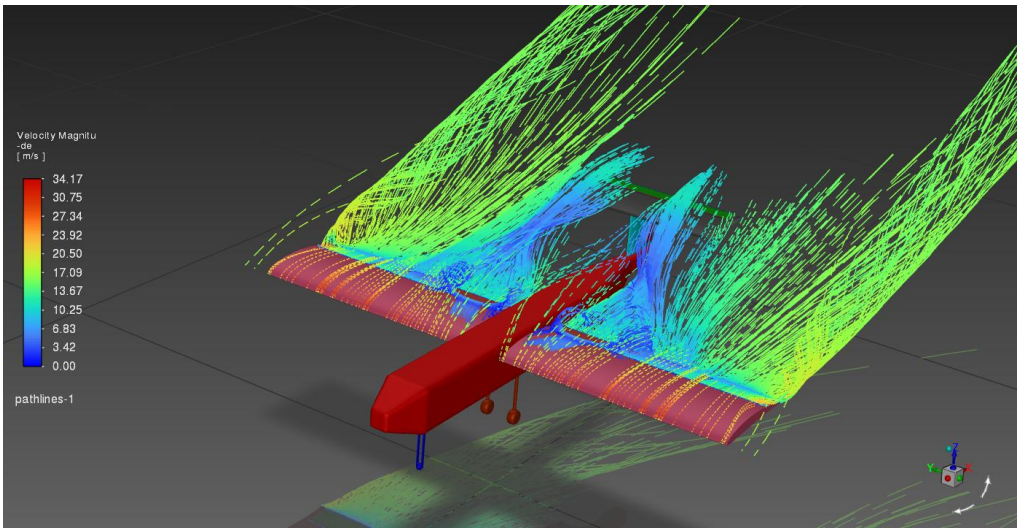


Figure 20:Post Stall Pathlines

At 18 degrees, the complex flow structure at the root breaks down and transitions to strong separation wake region, interacting with the elevator, causing loss of the lift on it as well, this explains the rapid loss of lift as the wing and tail stops generating lift.

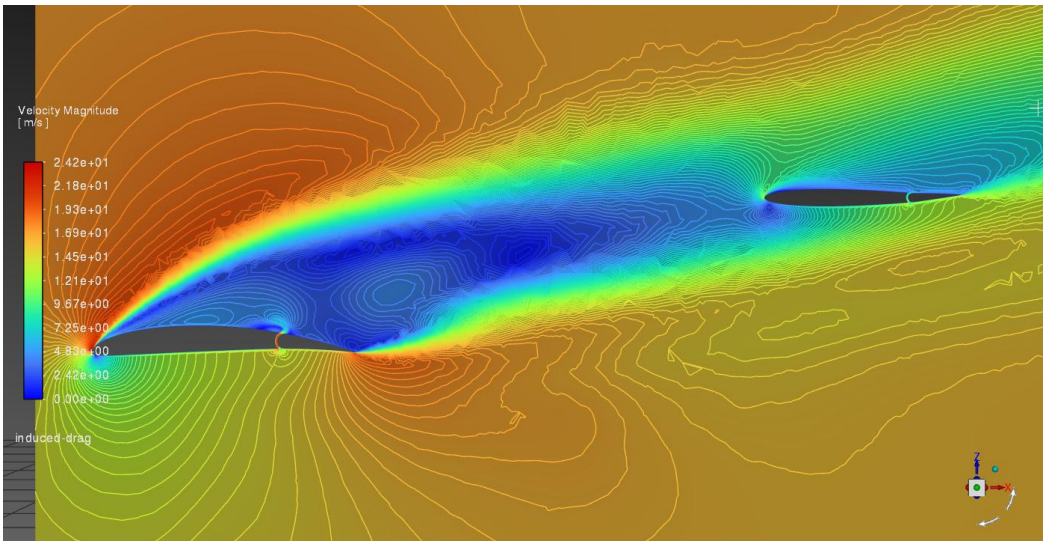


Figure 21:Velocity Contour at Root

Here we can clearly see the massive wake region caused by stalled root and interacting with the elevator, the contour lines show a recirculation zone just above the trailing edge, indicating a Von Karman Vortex Shedding.

Here's a **Video Animation** of the stalled wing pathlines.

Cruise

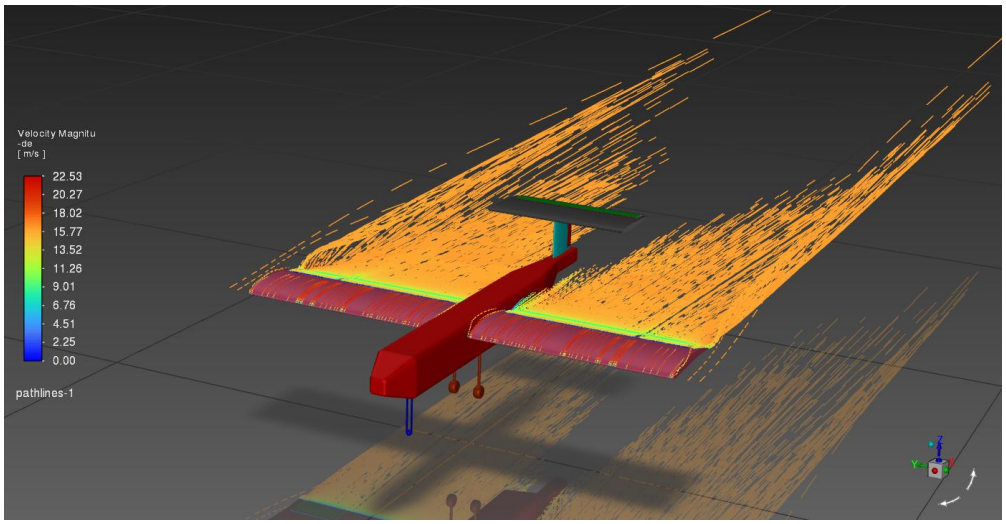


Figure 22: Pathlines of Cruise

Cruise shows attached flow with small tip vortex and gentle wake sheet curl up, indicating good efficiency and low induced drag due to lower lift coefficient.

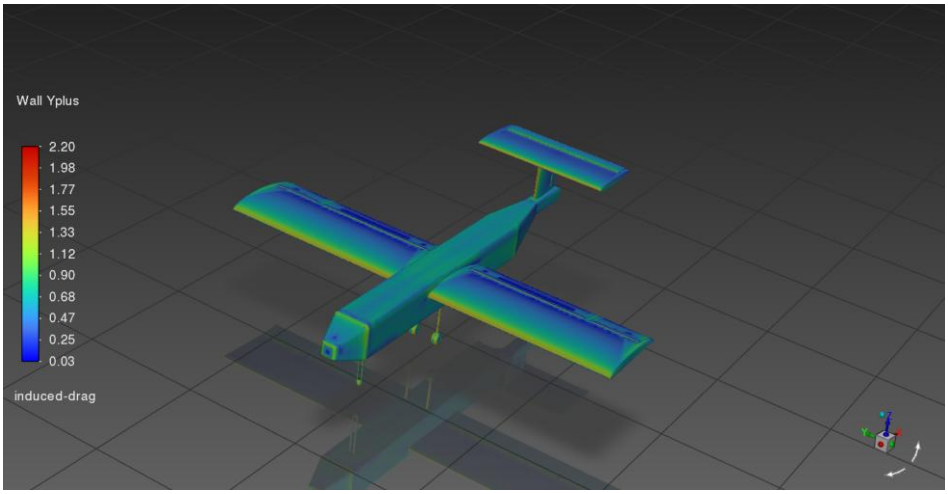


Figure 23: y+ of Cruise Speed

Lift, Drag and Pitching Moment Polar

1. Lift Curve

The UAV shows a maximum Lift Coefficient C_{Lmax}=1.65. given that the section maximum lift coefficient C_{lmax}=1.8, the decrease in lift coefficient is reasonable.

The slope of the lift curve is equal to

$$C_{L\alpha} = 0.0799$$

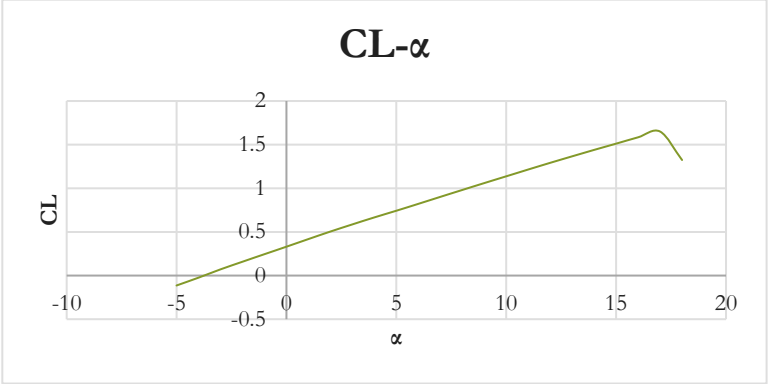


Figure 24:Lift Curve

2. Drag Polar

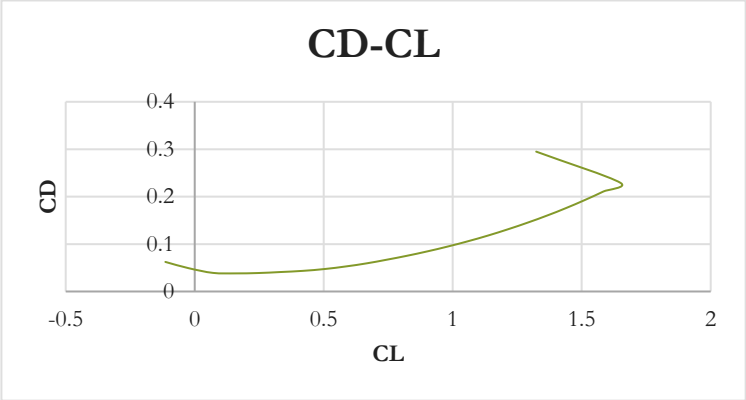


Figure 25:Drag Polar

•The drag polar features an asymmetric parabolic curve, with the drag after CL=0.079 being governed by a parabola.

$$C_D = 0.0874C_L^2 - 0.0317C_L + 0.0414$$

•Rewriting the equation in standard form

$$C_D = C_{Dmin} + K(C_L - C_{Ldmin})^2$$

$$\mathbf{C_D = 0.0874(C_L - 0.18135)^2 + 0.0385}$$

$$\because K = \frac{1}{\pi \cdot AR \cdot e}$$

$$\therefore \boldsymbol{e = 0.7283}$$

Which is a typical value for a straight rectangular wing UAV, further enhancements will be done to raise its value in the following project.

3. Pitching Moment

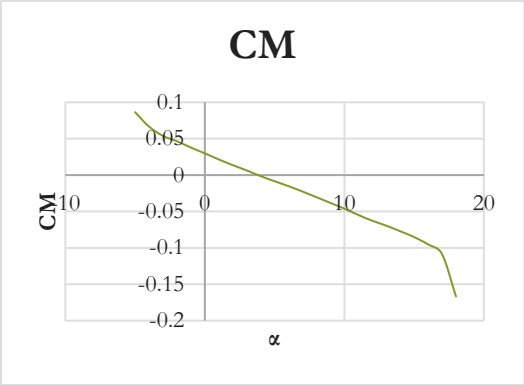


Figure 26:CM-a

•The Pitching Moment coefficient shows that the trim elevator deflection of -3.5 at α=2 was correctly estimated, the UAV is statically stable and the tail sizing was correct.

•The slope of the pitching moment curve is equal to

$$\boldsymbol{C_{m\alpha} = -0.0077}$$

$$\because Static\ Margin = -\frac{C_{m\alpha}}{C_{L\alpha}} \times 100$$

$$\therefore \boldsymbol{Static\ Margin = 9.637\%}$$

•Which is close from the estimated value of 14%, the UAV is statically stable.

•If we want to get the place of the neutral point X_{NP}

$$Static\ Margin = -\frac{X_{CG} - X_{NP}}{\bar{C}} \times 100$$

$$\therefore \boldsymbol{X_{NP} = 0.9065m}$$

4. CL-CD Curve

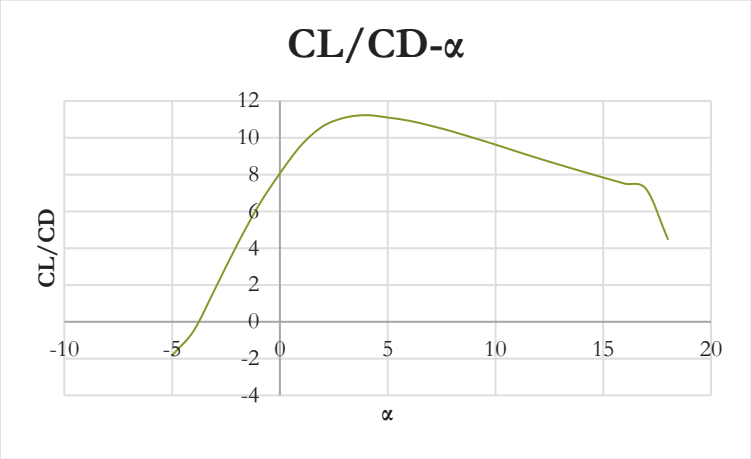


Figure 27:CL/CD-a

•The maximum CL/CD is exactly at 4 degrees, exactly as estimated. The maximum CL/CD is 11.23, the estimated value was 11.33, which is 0.178% away from the real value.

Spanwise Lift Distribution

•The industry standard of plotting the spanwise lift distribution is by splitting the wing into several sections, then plot the lift of each section with their central spanwise location.

•In this study, a different, more efficient approach was taken.

The following mesh parameters was extracted from the study:

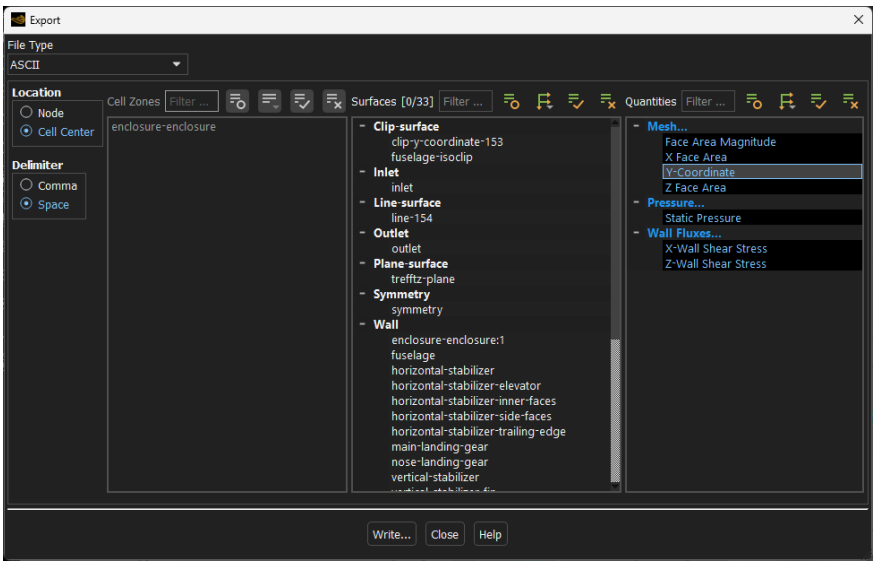


Figure 28:Parameters Required to Plot Spanwise Lift Distribution

Lift and Drag can be calculated the vectors at each mesh cell using the following equations inside Excel

$$l_{local} = \hat{p} * (-\overrightarrow{A_x} \cdot sin\alpha + \overrightarrow{A_z} \cdot cos\alpha)$$

$$d_{pressure} = \hat{p} * (\overrightarrow{A_x} \cdot cos\alpha + \overrightarrow{A_z} \cdot sin\alpha)$$

$$d_{viscous} = \bar{A} \cdot (\tau_x \cdot cos\alpha + \tau_z \cdot sin\alpha)$$

Then we can bin the components according to their spanwise location, add them together at the center of the range and plot them to get the spanwise lift distribution.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
y-coord	y-coord	press	x-face-	z-face-	x-wall-s	z-wall-s	face-area	local lift	local Drag	Local Visc	Bin Min(n	Bin Max(n	Bin Center	Lift_Local(I	Elliptical	Pressure	Viscous D	Total Dra	BM (NM)	Lift_t (N)	Total Dra	Total Visc	Total Dra	Step
-0.10517	1.05E-01	-1.39E+01	-1.89E-07	2.91E-07	-1.45E-01	-9.36E-02	3.59E-07	-4.428E-06	1.883E-06	-5.71E-08	-0.105	-0.159	-0.13175	2.528393	3.050	0.037999	0.019768	0.057767	22.79114	47.92	3.092356	0.422634	3.51499	0.0535
-0.10518	1.05E-01	-1.07E+01	-1.21E-07	-1.31E-07	-7.54E-02	6.36E-02	1.78E-07	1.158E-06	1.521E-06	-1.13E-08	-0.159	-0.212	-0.18525	2.569519	3.042	0.037172	0.016793	0.053965	20.35628					Pd
-0.10519	1.05E-01	6.36E+01	8.51E-08	3.26E-07	-1.02E+00	2.67E-01	3.37E-07	1.946E-05	8.929E-06	-3.24E-07	-0.212	-0.266	-0.23875	2.642777	3.027	0.067324	0.018484	0.085808	18.05863					48.6864
-0.10519	1.05E-01	7.16E+01	8.08E-08	3.40E-07	-6.53E-01	1.55E-01	3.49E-07	2.298E-05	9.926E-06	-2.15E-07	-0.266	-0.319	-0.29225	2.671097	3.004	0.09318	0.019176	0.112356	15.90223					
-0.10522	1.05E-01	-1.18E+01	-4.03E-07	1.03E-07	-5.78E-02	-1.93E-01	4.24E-07	-2.023E-06	4.470E-06	-3.84E-08	-0.319	-0.373	-0.34575	2.6963	2.972	0.106314	0.019658	0.125973	13.88869					
-0.10522	1.05E-01	-8.70E+00	7.27E-08	4.74E-07	4.97E-01	-7.62E-02	4.79E-07	-3.949E-06	-1.339E-06	2.28E-07	-0.373	-0.426	-0.39925	2.6918772	2.933	0.121651	0.019928	0.14158	12.01942					
-0.10522	1.05E-01	-2.14E+02	2.90E-07	-2.99E-07	7.22E-01	7.01E-01	4.16E-07	7.371E-05	-4.982E-05	3.47E-07	-0.426	-0.480	-0.45275	2.6699686	2.885	0.128417	0.019942	0.148358	10.29425					
-0.10522	1.05E-01	-2.10E+02	3.11E-07	-3.61E-07	7.47E-01	6.45E-01	4.76E-07	8.585E-05	-5.098E-05	4.04E-07	-0.480	-0.533	-0.50625	2.6471811	2.828	0.140859	0.020061	0.16092	8.712015					
-0.10522	1.05E-01	5.08E+01	1.19E-07	3.78E-07	-1.51E+00	4.74E-01	3.96E-07	1.785E-05	9.296E-06	-5.57E-07	-0.533	-0.587	-0.55975	2.6077249	2.761	0.140569	0.020095	0.160664	7.271567					
-0.10522	1.05E-01	-2.15E+02	3.10E-07	-3.01E-07	7.69E-01	7.95E-01	4.32E-07	7.533E-05	-5.435E-05	3.87E-07	-0.587	-0.640	-0.61325	2.5984671	2.685	0.150846	0.020307	0.171153	5.970776					
-0.10523	1.05E-01	7.97E+01	1.03E-07	5.17E-07	-2.46E-01	4.88E-02	5.27E-07	3.916E-05	1.523E-05	-1.23E-07	-0.640	-0.694	-0.66675	2.5291717	2.596	0.155354	0.02022	0.175573	4.809186					
-0.10523	1.05E-01	2.95E+01	1.59E-07	3.78E-07	-2.04E+00	8.51E-01	4.10E-07	1.017E-05	6.559E-06	-7.64E-07	-0.694	-0.747	-0.72025	2.4986427	2.496	0.155785	0.020513	0.176298	3.783112					
-0.10523	1.05E-01	-2.16E+02	3.29E-07	-3.00E-07	7.58E-01	8.36E-01	4.45E-07	7.606E-05	-5.872E-05	3.97E-07	-0.747	-0.801	-0.77375	2.4110627	2.381	0.16553	0.020454	0.185984	2.890892					
-0.10523	1.05E-01	-2.14E+02	4.05E-07	-1.91E-07	6.80E-01	1.45E+00	4.48E-07	5.552E-05	-7.832E-05	4.13E-07	-0.801	-0.854	-0.82725	2.3434642	2.251	0.17283	0.023129	0.195959	2.127903					
-0.10523	1.05E-01	-2.07E+02	4.19E-07	-1.57E-07	6.33E-01	1.71E+00	4.47E-07	4.707E-05	-7.964E-05	4.11E-07	-0.854	-0.908	-0.88075	2.2450703	2.101	0.178391	0.022916	0.201307	1.490584					
-0.10523	1.05E-01	-2.19E+02	3.89E-07	-2.19E-07	6.90E-01	1.24E+00	4.46E-07	6.201E-05	-7.536E-05	3.99E-07	-0.908	-0.961	-0.93425	2.1754301	1.928	0.181371	0.021007	0.202378	0.973704					
-0.10523	1.05E-01	-2.20E+02	3.77E-07	-2.43E-07	6.60E-01	1.03E+00	4.48E-07	6.703E-05	-7.227E-05	3.72E-07	-0.961	-1.015	-0.98775	2.0405195	1.724	0.194247	0.021159	0.215407	0.573485					
-0.10523	1.05E-01	-1.96E+02	4.28E-07	-1.19E-07	4.91E-01	1.78E+00	4.44E-07	3.762E-05	-7.848E-05	3.52E-07	-1.015	-1.068	-1.04125	1.9119994	1.477	0.209648	0.021678	0.231326	0.282796					

Figure 29: Spanwise lift and drag post processing in excel

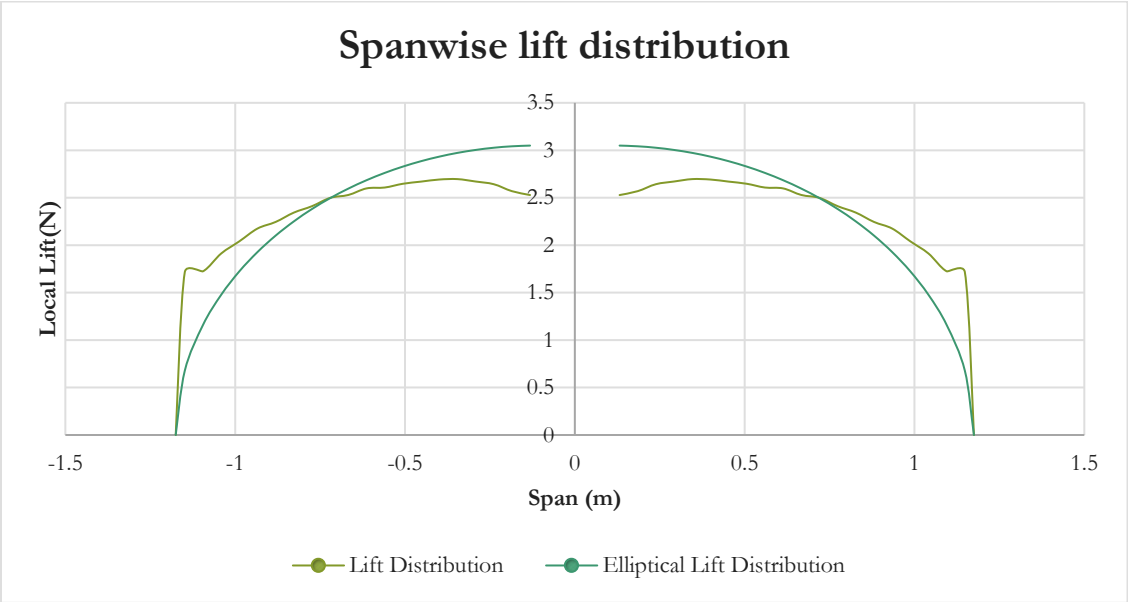


Figure 30:Lift Distribution

The lift distribution shows a typical behavior of a rectangular wing. The non-elliptical wing loading creates excessive lift near wingtips, this increases the recirculation at the wingtips causing strong tip vortices, its effect on the drag can be seen in the following graph, with drag spiking near wingtips due to the tip vortices.

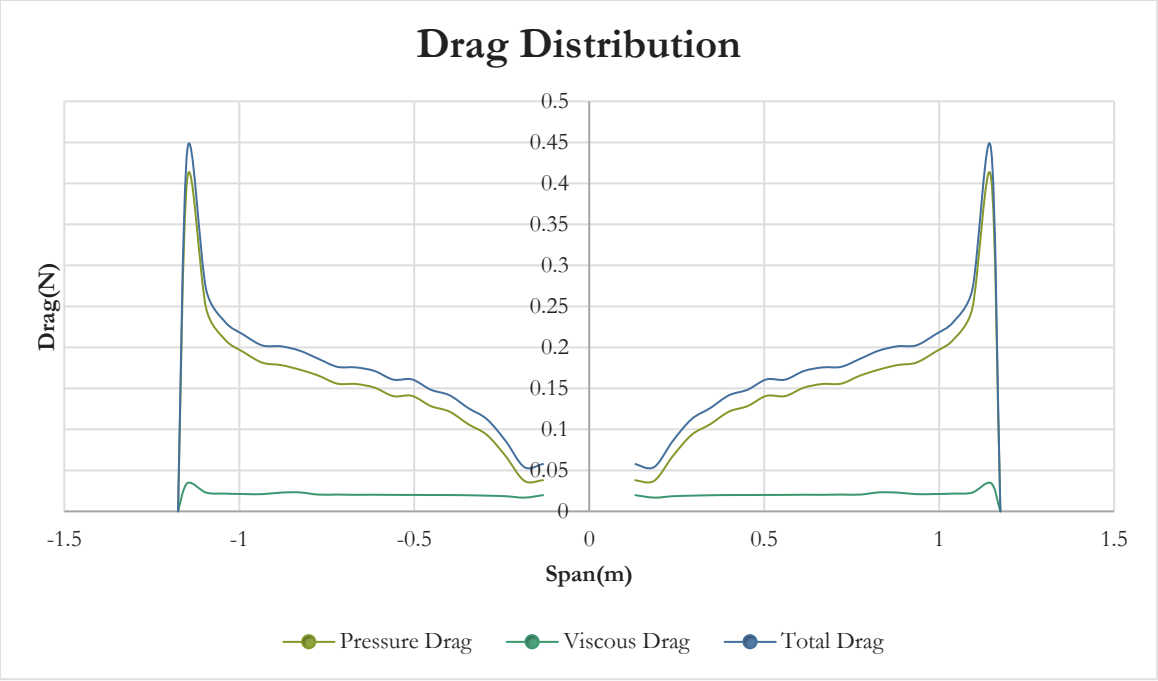


Figure 31:Drag Distribution Across Span

Summary of Key Aerodynamic Factor

Parameter	CFD	Estimated/VLM	Estimation Error
C_{Lmax}	1.654	VLM can't predict it	-
$\frac{C_L}{C_{D\max}}$	11.23	11.33	0.178%
$\alpha_{\frac{C_L}{C_{D\max}}}$	4	4	0
e	0.7283	VLM can't predict it	-
Static Margin	9.67%	14.25%	47.36%
X_{NP}	0.9053	0.926	-2.29%
C_{Dmin}	0.0385	0.0377	-2.078%